Curriculum Vitae

WORK EXPERIENCE

Research Fellow in Extragalactic Astronomy 01/04/2024 – Present
Astronomy Centre, University of Sussex, United Kingdom

DAWN Fellow 15/10/ 2021 – 29/02/2024
Cosmic Dawn Center, DTU Space, Technical University of Denmark (DTU)

EDUCATION

PhD in Astronomy 30/09/2017 – 06/12/2021
Astronomy Centre, University of Sussex, United Kingdom
Advisors: Prof. Peter A. Thomas (main supervisor) & Prof. Stephen M. Wilkins (second supervisor)
Title: The effect of dust in observing galaxies in the early Universe
PhD Viva: 04/10/2021; examiners: Prof. Romeel Davé & Prof. Seb Oliver, PhD Award date: 06/12/2021

MSc. Research in Astronomy, Cosmology 01/09/2015 – 31/08/2017
Sterrewacht Leiden, Universiteit Leiden, The Netherlands
Major Project: Analysing the impact of environment and mergers on halo concentrations, supervisor Dr. Camila A. Correa.
Minor project: A search for serendipitous emission lines in ALMA observations of high-redshift galaxies, supervisor Dr. Jacqueline Hodge.

(Took a one year break to fill knowledge gaps in order to switch fields from engineering to astrophysics, also apply to graduate schools)

Bachelor of Technology (B.Tech) in Engineering Physics 01/08/2010 – 31/07/2014
National Institute of Technology Calicut (NITC), Kerala, India
Thesis: One Dimensional Organic Semiconductor Nanostructures for Solar Cell Fabrication, supervisor Dr. Vari Sivaji Reddy

TEACHING & OTHER EXPERIENCE

Associate Tutor, School of Mathematical & Physical Sciences, University of Sussex 2025-2026
Associate Tutor for the Foundation year physics course Properties of Matter (F3216)
Associate Tutor for undergraduate physics course Heat and Modern Physics (F3251)

Lecturer, School of Mathematical & Physical Sciences, University of Sussex Jan-May 2025
Lecturer for the master's course Astrophysical Processes

Lecturer on SED fitting and forward modelling galaxies at ACES PhD Summer School, Sussex Sep 2024

Supervision of Ruru Thakur, PhD student University of Sussex, UK
co-supervision with Prof. Stephen M. Wilkins 2025 - Present

Supervision of Jack Turner, PhD student University of Sussex, UK
co-supervision with Prof. Stephen M. Wilkins 2023 - Present

Supervision of Malte Brinch, PhD student Cosmic Dawn Center, DTU
co-supervision with Prof. Thomas R. Greve 2022 - 2024

Supervision of Paurush Punyasheel, MSc student from BITS Goa, India Cosmic Dawn Center, DTU
Panchromatic galaxy sizes in FLARES Sep-May 2023/24

Supervision of Rebeca G Reyes Carrion, Dawn-IRES student (University of Peurto Rico, USA) Star formation efficiency in FLARES Cosmic Dawn Center, DTU June-Aug 2022

Co-supervision of Soren Staal, BSc student Cosmic Dawn Center, DTU
Analysing the morphology of high-z galaxies with FLARES and JWST Feb-June 2022

Co-supervision of Hamish Garnett, Mphys student University of Sussex
Mass and metallicity gradients in FLARES, main supervisor Prof. Peter A. Thomas Oct-Apr 2019/20
Effect of random seeds and AGN on galaxy properties in the EoR using EAGLE simulation physics, main supervisor Prof. Peter A. Thomas Oct-Apr 2020/21

PhD Work Placement, with Kenya Red Cross Society, Kenya (Online) Mar-Nov 2020

Validation of inferences from satellite observations with ground survey data for drought impact studies

Associate Tutor, School of Mathematical & Physical Sciences, University of Sussex 2018-2020
Associate Tutor for the Foundation year physics course Properties of Matter (F3216)
Associate Tutor for the third year undergraduate physics course Extragalactic Astronomy (F3209)

PROFESSIONAL ACTIVITIES & ORGANISATION EXPERIENCE

co-PI of the SYNTHESIZER Project, a fast and flexible tool for synthetic observations Apr 2022-Present

co-PI of the FLARES Project, suite of re-simulations specially designed to study galaxies observable in the high-redshift Universe Jan 2020-Present

Co-organiser of Sussex Astronomy Centre weekly colloquium Sep 2024 - Present

Main Organiser of International High-Redshift Workshop in Kerala, India 5-9 January 2026

Reviewer for Monthly Notices of the Royal Astronomical Society (MNRAS), and Astronomy & Astrophysics (A&A) Oct 2021 – Present

Session Chair, High-Redshift galaxies: Apples to oranges from dusk to dawn, European Astronomical Society (EAS) Meeting 2024, Padova, Italy July 2024

Session co-Chair, **Dust enrichment of early galaxies (z>5) in the era of JWST and ALMA** and **Removing the Disguise: SMGs in the era of JWST**, EAS Meeting 2024 July 2024

Astronomy on Tap (AoT), Copenhagen, Organising committee member June 2022 – Dec 2023

Co-organiser of DAWN caketalks, Weekly talks at the Cosmic Dawn Center Oct 2021 – Sep 2023

2023 DTU Astronomy PhD Selection Committee Member June-July 2023
Committee of peers to select PhD students for astronomy at DTU Space

2022 DAWN Fellowship Committee Member Jan-Feb 2022
Committee of peers to select from applicants for the 2022 DAWN Postdoctoral Fellowship

SOC, Models and Simulations of High-Redshift Galaxies - Sazerac Sip Oct 2021
A short online conference to discuss current models and simulations of high-redshift galaxies

SOC, CIDER: Cold ISM During the Epoch of Deionisation - Sazerac Sip Feb 2021
A short online conference to discuss current works on the cold ISM and dust in the EoR

OUTREACH

Goodwood Festival of Speed, UK; Talked to the public about the Euclid Telescope July 2025, 2026

Lewes STEM Festival, UK; Talked to the public about spectroscopy in astronomy Dec 2025

Astronomy on Tap (AoT), Copenhagen, Denmark Feb 2021
Talked about cosmological simulations of galaxy formation and evolution

BlueDot Festival, Manchester, UK July 2018
Volunteered at the Webb Telescope stand

ACHIEVEMENTS

PI of DiRAC computing time to forward model galaxy simulations and set-up spectral emulators (1.36 Mcore-h of CPU time and 10 KGPU-h) 2026 - 2027

Marie Skłodowska-Curie Actions Postdoctoral Fellowships, Seal of Excellence 2024

DAWN Postdoctoral Fellowship, Cosmic Dawn Center, DTU, Denmark 15/10/2021 – 29/02/2024

Awarded Millions of CPU hours on DiRAC/COSMA through Virgo Consortium membership 2018 – Present

co-PI/co-I on several successful ALMA and JWST proposals to study the early Universe 2021 – Present

DISCnet scholarship to pursue PhD at University of Sussex, UK 30/09/2017 – 30/09/2021

Partial **Oort/NOVA MSc scholarship** from Leiden Observatory, The Netherlands 01/09/2015 – 31/07/2021

GATE (Graduate Aptitude Test) and JEST (Joint Entrance Screening Test) in Physics, 2015
All India Rank 88 and 71 (Some of the most competitive exams for graduate schools in India)

NET (National Eligibility Test, for Junior Research Fellowships) in the Physical Sciences, Dec 2014
All India Rank 51